MaatriCare — Product Requirements Document

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1. Overview

MaatriCare+ is an AI-assisted platform for rural public health. It helps ASHA workers and PHC doctors catch high-risk pregnancies early, and tracks every referral until the patient is actually seen. It is built for the Smart India Hackathon 2026 problem statement 26133: *Accessibility and quality of public healthcare services, particularly in rural and underserved areas* (MedTech / HealthTech, Software).

Problem

In rural India, maternal and child health risks are often caught too late. Diagnostics are hard to reach, doctors are scarce, and Sub-Centres, PHCs and district hospitals coordinate poorly. When a patient is handed from one facility to the next, the referral and the record often get lost along the way. Low connectivity makes all of this worse.

Existing public tools each solve one piece of the problem:

Capability	MaatriCare+	eSanjeevani	108/104 helpline	ABHA/ABDM portal
AI risk prediction	Yes	No	No	No
Teleconsultation	Yes	Yes	No	No
Appointment / queue management	Yes	Partial	No	No
Offline-first ASHA app	Yes	No	No	No
Multi-tier referral tracking	Yes	No	Partial	No
Emergency escalation	Yes	No	Yes	No
Diagnostic / medicine-stock visibility	Yes	No	No	No
FHIR / ABDM compliance	Yes	Partial	No	Yes
Tamper-proof audit trail	Yes	No	No	No

Capability	MaatriCare+	eSanjeevani	108/104 helpline	ABHA/ABDM portal
Multilingual / voice support	Yes	Partial	Yes	No

Product vision

Take rural care from prediction to action, with accountability. Every at-risk mother and child is flagged early. Each one is connected to the right level of care and followed until the case is closed. Every step leaves an auditable record.

What MaatriCare+ is not

It does not replace eSanjeevani, 108 or ABDM. It plugs into them. It adds a risk-and-referral layer the public system lacks today.

2. Goals, non-goals and success metrics

Goals

1. **Predict risk early.** Score maternal, child and chronic risk from vitals (BP, Hb, BMI, obstetric history), so intervention starts before an emergency.
2. **Bring the doctor to the village.** Run ASHA-assisted video or audio consults with PHC doctors, so patients travel less.
3. **Lose no patient between facilities.** Track every referral from Sub-Centre to PHC, Rural Hospital and District Hospital until it is closed.
4. **Work without internet.** ASHA workers can capture data, triage and book care offline, then sync later.
5. **Make accountability tamper-proof.** Write referrals, escalations and outcomes to an immutable audit log.
6. **Fit national standards.** Use FHIR and ABDM so records move into India's digital health stack rather than creating a new silo.

Non-goals

- Making autonomous diagnoses or prescriptions. The AI flags risk; a clinician decides.
- Replacing eSanjeevani, 108/104 or the ABHA/ABDM portal.
- Billing, insurance claims or hospital management (HMIS) features.

- Putting patient health data on a blockchain. Only hashes and event metadata go on-chain.

Success metrics

The idea document names three evaluation metrics: referral completion rate, response time and risk-detection accuracy. The targets below are proposed starting points. They should be baselined and confirmed during the pilot.

Metric	Definition	Proposed pilot target
Referral completion rate	Referrals marked "seen at destination" ÷ referrals created, within 7 days	⩾ 80%
Emergency response time	Time from high-risk alert to 108 dispatch confirmation	< 10 min
Risk-detection sensitivity	High-risk cases correctly flagged ÷ all true high-risk cases (validation set)	⩾ 85%
False-positive rate	Low-risk cases flagged high-risk ÷ all low-risk cases	⩽ 20%
ANC coverage	Registered pregnancies with 4+ antenatal visits recorded	+15 pts over baseline
ASHA weekly active use	ASHA workers who log at least one visit per week	⩾ 70%
Offline sync success	Offline records synced without conflict or loss	⩾ 99%

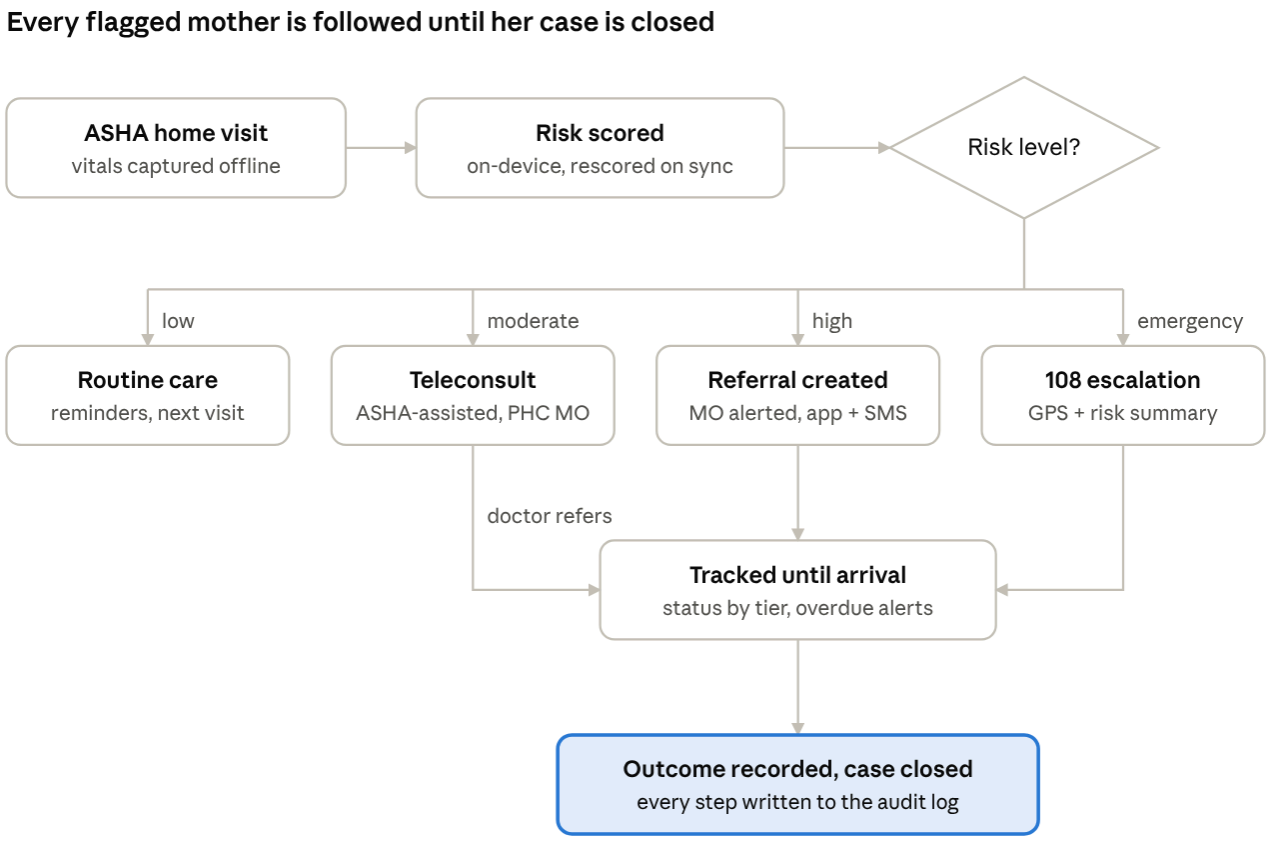
3. Target users and personas

The ASHA worker is the primary user. Every other role depends on the data she captures in the field.

Persona	Where they work	What they need from MaatriCare+	Primary surface
ASHA worker (primary)	Village, home visits, Sub-Centre	Register pregnancies and children, record vitals offline, see a risk score, book consults and referrals, follow up on due visits	Offline-first Android app (Flutter)
Pregnant woman / mother and family	Home, village	Reminders, health guidance in her language, a consult without travelling, help fast in an emergency	SMS, IVR, multilingual chatbot / voice assistant
PHC medical officer	Primary Health Centre	A ranked list of high-risk patients, explanations for each flag, teleconsults, referrals out and in, today's queue	Doctor web dashboard
ANM / Sub-Centre staff	Sub-Centre	Supervise ASHA entries, check vitals, start referrals	Mobile app + web dashboard
Rural / District Hospital doctor	CHC, Rural or District Hospital	Advance notice of incoming referrals with a risk summary, a way to confirm arrival and record the outcome	Doctor web dashboard
Block / district health administrator	Block or district office	Village-level risk heatmaps, referral success rates, facility load, medicine stock, audit reports	Admin web dashboard
108 EMS dispatcher	State emergency control room	A pre-filled patient risk summary with GPS location when an escalation fires	108 EMS API integration

4. User journeys

Everything in MaatriCare+ supports one loop: an ASHA visit produces a risk score, and the score routes the patient to the right level of care. Every high-risk case is then tracked until it is closed.



core care journey · 4 risk paths, 1 closed loop

Low-risk patients go back into the routine visit cycle. Any referral that misses its arrival window alerts the referring ASHA and the PHC again.

Key use cases

1. **Field registration, offline.** In a village with no signal, an ASHA registers a 24-week pregnant woman, records BP and Hb, and sees a provisional High score with the reasons (e.g. low Hb, previous C-section). The record syncs when she reaches coverage.
2. **Doctor triage.** Each morning, the PHC medical officer opens a worklist ranked by risk. They confirm or override each new flag with a reason.
3. **Assisted teleconsult.** For a Moderate case, the ASHA starts a video call to the PHC doctor from the patient's home. When bandwidth drops, the call falls back to audio. Advice and prescriptions save to the record.

4. **Referral with handoff.** The MO refers a High-risk patient to the District Hospital. The hospital sees her risk summary before she arrives, marks her Arrived, and records the outcome. The case then closes.
5. **Emergency.** The ASHA spots heavy bleeding and taps Emergency. 108 receives GPS and a pre-filled risk summary. The PHC and district hospital are notified at the same time.
6. **Oversight.** A block health officer reviews the village risk heatmap and referral completion by facility. They export the monthly report and audit any disputed referral against its ledger hash.
7. **Patient self-help.** A mother asks the chatbot in her own language about danger signs. She gets her next ANC reminder by SMS.

5. Functional requirements

The MVP (P0) is the core loop: an offline ASHA app, maternal risk scoring, a basic chatbot and referral tracking. P1 adds teleconsultation, 108 escalation and admin analytics. P2 adds blockchain anchoring, diagnostics and stock visibility, and chronic-care scoring. This follows the phased rollout in the idea document.

Priority key: P0 = MVP / pilot, P1 = Phase 2, P2 = Phase 3.

ID	Module	Requirement	Priority
FR-01	Registration & records	ASHA registers a pregnant woman, child or chronic patient and links an existing ABHA ID, or creates one later when online.	P0
FR-02	Registration & records	Capture vitals and history: BP, Hb, weight/height (BMI), gestational age, obstetric history, danger signs. Input is validated against plausible ranges.	P0
FR-03	Registration & records	Each patient has a longitudinal record (visits, vitals, scores, referrals, consults) stored as FHIR R4 resources.	P0
FR-04	Offline-first	All capture, triage, booking and referral creation works with no connectivity. Data is stored in an encrypted local SQLite database.	P0
FR-05	Offline-first	Background sync runs when a connection appears. Conflicts are resolved on the server, and each record shows its sync status.	P0

ID	Module	Requirement	Priority
FR-06	Offline-first	A lightweight on-device model gives a provisional risk score offline. The server rescores after sync.	P1
FR-07	AI risk prediction	Maternal risk score (Low / Moderate / High) from vitals and history, using Logistic Regression or XGBoost.	P0
FR-08	AI risk prediction	Child risk score (e.g. malnutrition, missed immunisation, low birth weight).	P1
FR-09	AI risk prediction	Chronic risk score (e.g. hypertension, diabetes, anaemia).	P2
FR-10	AI risk prediction	Every score shows its top contributing factors in plain language (explainable AI).	P0
FR-11	AI risk prediction	Configurable clinical danger-sign rules always force a High flag, whatever the model outputs.	P0
FR-12	AI risk prediction	A doctor can confirm or override a score with a reason. Overrides are logged and fed into retraining.	P1
FR-13	Alerts & escalation	A High flag alerts the PHC medical officer (dashboard plus SMS via Twilio or MSG91) and the assigned ASHA.	P0
FR-14	Alerts & escalation	An alert not acknowledged within a configurable time escalates to the next responsible person.	P1
FR-15	Alerts & escalation	One-tap emergency sends the State 108 EMS a request with GPS location and a pre-filled risk summary.	P1
FR-16	Teleconsultation	ASHA-assisted video consult between patient and PHC doctor (WebRTC / Agora SDK).	P1
FR-17	Teleconsultation	Calls fall back to audio automatically on low bandwidth (Twilio Voice).	P1
FR-18	Teleconsultation	Consult notes, advice and e-prescription are saved to the patient record and visible to the ASHA.	P1

ID	Module	Requirement	Priority
FR-19	Referral & resources	Create a referral to a higher tier (Sub-Centre, PHC, Rural Hospital, District Hospital) carrying the reason, risk summary and records.	P0
FR-20	Referral & resources	Referral lifecycle: Created, Accepted, In transit, Arrived, Outcome recorded, Closed. Each change is timestamped with who made it.	P0
FR-21	Referral & resources	Referrals not marked Arrived within a set window alert the referring ASHA and the PHC for follow-up.	P0
FR-22	Referral & resources	Live lab-slot visibility and booking at the destination facility.	P2
FR-23	Referral & resources	PHC medicine-stock visibility for key maternal and child drugs, with low-stock alerts.	P2
FR-24	Appointments & queue	Book PHC and ANC appointments, with due-visit reminders to the patient and ASHA.	P1
FR-25	Appointments & queue	Live PHC queue with token status sent by SMS.	P2
FR-26	Patient engagement	Multilingual chatbot for health guidance, reminders and FAQs (Dialogflow / Python NLP).	P0
FR-27	Patient engagement	Voice assistant and IVR access for feature-phone and low-literacy users (speech APIs).	P2
FR-28	Dashboards	Doctor dashboard: risk-ranked worklist, patient timeline, score explanations, incoming and outgoing referrals.	P0
FR-29	Dashboards	Admin dashboard: village-level risk heatmap, referral success rate, facility load.	P1
FR-30	Dashboards	Export reports in formats needed for government programme reporting.	P2
FR-31	Audit & accountability	Append-only audit log of every clinical action: score, override, referral change, escalation.	P0

ID	Module	Requirement	Priority
FR-32	Audit & accountability	Hashes of record versions, referral milestones and emergency triggers are anchored to a permissioned blockchain.	P2
FR-33	Audit & accountability	An auditor can check any record or referral against its on-chain hash.	P2
FR-34	Interoperability & access	FHIR R4 data model and API (HAPI FHIR server) for all clinical data.	P0
FR-35	Interoperability & access	ABDM integration: ABHA linking and consent-based record sharing (HIP / HIU flows).	P1
FR-36	Interoperability & access	Role-based access scoped by role and facility (ASHA, ANM, MO, hospital doctor, admin, auditor).	P0

6. Non-functional requirements

Clinical safety and patient privacy come first. Every other requirement gives way to them.

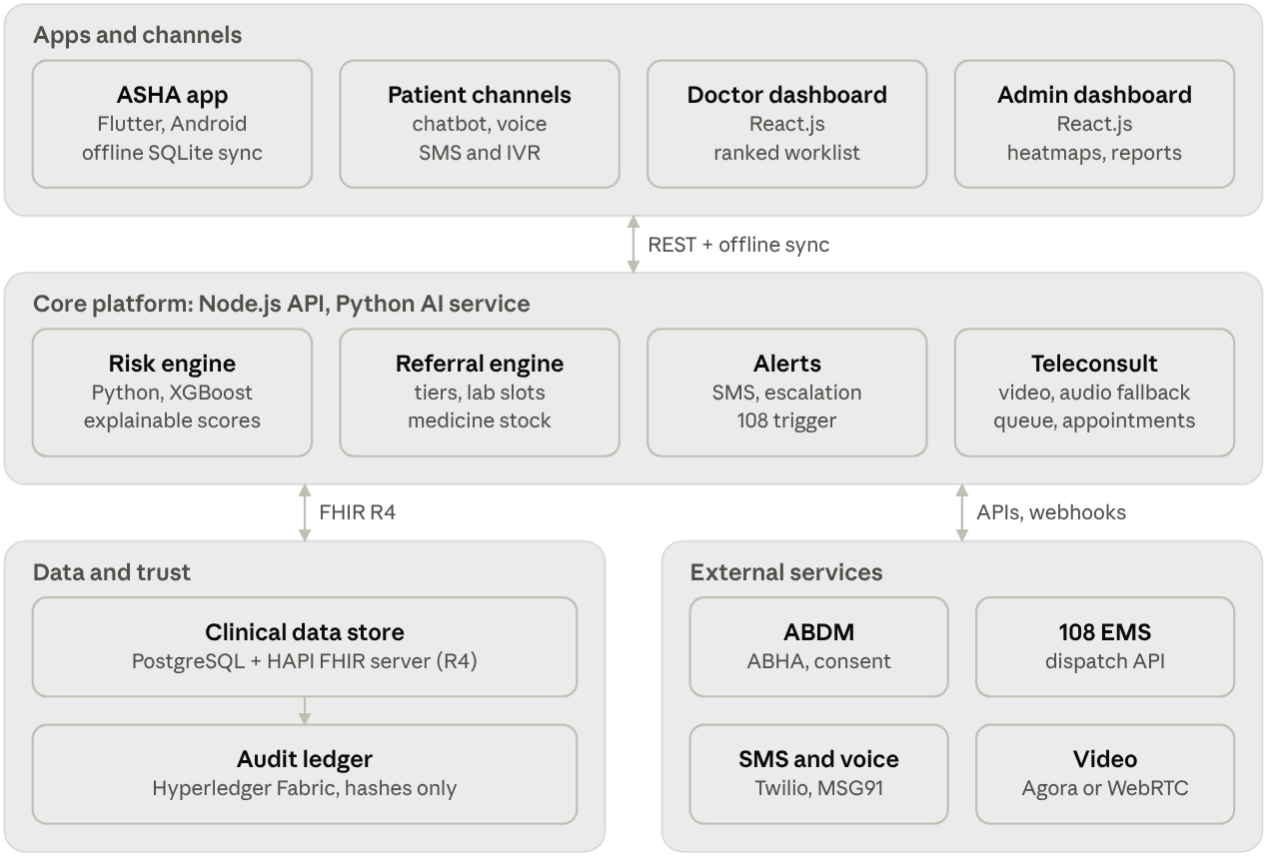
Area	Requirement
Clinical safety	AI output is decision support only. A clinician makes every treatment or referral decision. Danger-sign rules override the model (FR-11). Each model version is validated on a held-out set before release, and its sensitivity and false-positive rate are published to admins.
Privacy & consent	Comply with the Digital Personal Data Protection Act 2023 and ABDM consent rules. Collect explicit, recorded consent at registration, in the patient's language. Collect only the minimum data needed.
Data on blockchain	No personal or health data on-chain. The ledger holds only hashes, pseudonymous IDs, event types and timestamps.
Security	TLS 1.2+ in transit. AES-256 at rest, including the on-device SQLite store. Role- and facility-scoped access (FR-36). Every read of a patient record is logged. Remote wipe for lost devices.

Area	Requirement
Offline & sync	ASHA app works fully offline for at least 7 days of field work. Sync never silently drops or overwrites data. Conflicts are flagged for review.
Low-end devices	Runs on entry-level Android phones (Android 8+, 2 GB RAM). App install size kept small. Usable on 2G networks, with audio fallback for calls.
Performance	Risk score returned within 2 s online and immediately on-device offline. Dashboards load within 3 s on a typical PHC connection.
Availability	Backend targets 99.5% uptime. Alerting and 108 escalation paths have SMS fallback when the data network fails.
Localisation	UI, chatbot and voice in Hindi, English and at least two regional languages for the pilot state. Icon-led UI for low-literacy users.
Accessibility	Large touch targets, high contrast, audio prompts. Dashboards meet WCAG 2.1 AA.
Interoperability	FHIR R4, aligned with the ABDM / NRCES India FHIR implementation guide. Standard codes where available (e.g. LOINC for vitals, ICD-10 for diagnoses).
Scalability	Starts as a single-district pilot. The architecture scales horizontally to a state without redesign.
Explainability & audit	Every score stores its model version, inputs and top factors, so any flag can be explained after the fact.

7. System architecture and tech approach

All clients talk to one API core, and every clinical record is stored as FHIR. The blockchain holds only hashes, so patient data never leaves the governed database.

All apps share one FHIR-based core; only hashes reach the ledger



system architecture · 4 groups, 14 components

The ASHA app keeps a full local copy of its patients and syncs through the same API as the dashboards. The ledger receives a hash of each record version and referral event, never the record itself.

The idea document groups these parts into three layers. **Patient and community interaction** covers the apps, chatbot, voice and IVR. **Healthcare intelligence and coordination** covers the risk, referral, alert and teleconsult services. **Blockchain trust and accountability** is the audit ledger.

Tech stack

Layer	Choice	Notes
ASHA mobile app	Flutter, SQLite	Offline-first, encrypted local store, background sync
Web dashboards	React.js	Doctor and admin views, heatmaps

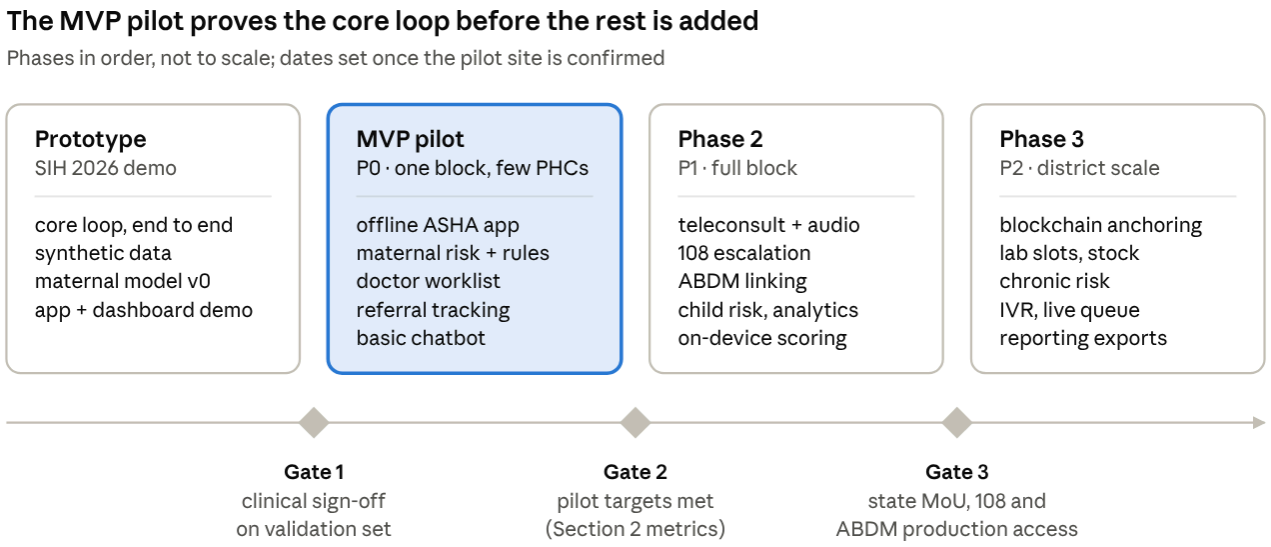
Layer	Choice	Notes
API backend	Node.js	REST API, auth, role-based access, sync endpoint
AI / ML	Python, scikit-learn, XGBoost (Logistic Regression baseline)	Served as a separate service; SHAP-style factor explanations; small exported model for on-device scoring
Clinical data	PostgreSQL + HAPI FHIR server	FHIR R4, aligned with the ABDM / NRCES implementation guide
Chatbot & voice	Dialogflow or Python NLP, speech APIs	Multilingual; IVR for feature phones
Teleconsultation	WebRTC / Agora SDK, Twilio Voice fallback	Audio-only mode on weak networks
Messaging	Twilio or MSG91	SMS alerts, reminders, OTP
Emergency	State 108 EMS API	GPS plus pre-filled risk summary
Audit ledger	Hyperledger Fabric (recommended) or Ethereum	Permissioned network, hashes only

Key technical decisions

- **Hyperledger Fabric over public Ethereum.** Government health data needs a permissioned network with known participants and no gas fees. This is still listed as an open question.
- **Rules plus model.** Clinical danger-sign rules run before the ML score and can force a High flag. This keeps safety-critical logic deterministic and easy to audit.
- **Server is the source of truth.** The device score is provisional. The server rescores after sync, and any change of risk level triggers the normal alert flow.

8. Release plan

The rollout is pilot-first, as the idea document proposes. The MVP pilot (AI model, chatbot and ASHA app) must prove the core loop before blockchain, teleconsultation and diagnostics are added.



phased roadmap · 4 phases, 3 gates

Each phase delivers the requirements tagged with its priority in Section 5. Gate 1 matters most: no risk score reaches a real patient until clinical advisers sign off on the model's validation results.

Hackathon demo scope

- [ ] ASHA app: register a patient, enter vitals offline, see a risk score with its reasons
- [ ] Risk engine: maternal model trained on synthetic and public data, plus danger-sign rules
- [ ] Doctor dashboard: risk-ranked worklist, create a referral, follow its status to Closed
- [ ] Chatbot: answer danger-sign questions in two languages
- [ ] Admin view: village risk heatmap on seeded data
- [ ] Audit: show a referral's history and its hash written to a test ledger

9. Risks, assumptions, dependencies and open questions

Risks

Risk	Impact	Mitigation
Little real rural clinical data	Model may not reflect real patients	Train on synthetic data plus public datasets; validate with a partner PHC before any clinical use; retrain on pilot data

Risk	Impact	Mitigation
False negatives / false positives	Missed high-risk mothers, or alert fatigue	Danger-sign rules override the model; tune for sensitivity; doctor override and feedback loop; track both rates every release
Poor connectivity and few devices	Data gaps, delayed alerts	Offline-first app, SMS and IVR fallback, audio-only consults
Health workers distrust AI	Low adoption	Explainable scores, ASHA and doctor training, AI framed as an assistant, not a decision-maker
Multi-tier referral coordination	Referrals stall between facilities	Clear status lifecycle, overdue alerts, named owner at each tier
Integration access (108 EMS, ABDM)	Escalation or record sharing blocked	Start with ABDM sandbox and SMS-based escalation; pursue state MoUs early
Setup cost (devices, training)	Slow rollout	Pilot in one block first; reuse existing ASHA phones where possible
Regulatory classification of AI risk scoring	Approval needed before clinical deployment	Position as decision support; get legal review of medical-device rules before scaling

Assumptions

- ASHA workers have, or can be given, an Android smartphone.
- A district health office will sponsor a pilot and provide access to a PHC and its referral facilities.
- Patients or families can receive SMS on at least a basic phone.

Dependencies

- ABDM sandbox and production onboarding (ABHA, HIP / HIU registration).
- State 108 EMS API access.
- SMS and voice providers (Twilio or MSG91), video SDK (Agora) or self-hosted WebRTC infrastructure.

- Clinical advisers (obstetrician, paediatrician) to define danger-sign rules and validate models.

Open questions

- [ ] Ethereum or Hyperledger Fabric? A permissioned network (Fabric) suits government data and has no gas fees. Recommend Fabric.
- [ ] Which pilot state and district? This decides the languages, 108 integration path and facility list.
- [ ] Which datasets will train the first maternal model, and who signs off on clinical validation?
- [ ] Is chronic-care scoring in scope for the hackathon demo, or post-hackathon only?
- [ ] Should the patient-facing chatbot be a separate app, WhatsApp, or SMS / IVR only?
- [ ] What is the agreed response-time target with the 108 control room?

References

- eSanjeevani, National Teleconsultation Service
- Ayushman Bharat Digital Mission (ABDM): standards and ABHA
- HL7 FHIR India Implementation Guide (NRCES)
- National Health Mission: maternal health programmes
- WHO: maternal health